Intellectual Timeline of SURA Research

From adaptive regret to optimization-selected uncertainty quantification

Nicholas Mino | April 27–July 29, 2026 | **Reconstructed effort: 92 hours (June 1–July 29)**

Methodological note. Intellectual timeline, not a timesheet. Hours are conservative reconstructions from meetings, emails, drafts, repository history, papers, catalogs, and proofs. Weakly documented intervals are labeled; no undocumented mathematical pivots are invented.

Executive summary

Research evolution

June 29 meeting and July 2 written feedback drove the pivot to Formulation II: uncertainty sets whose hyperparameter is selected by constrained optimization on the same calibration data.

Milestone status

Milestone	Status
Formulation I	Completed
Stabilized-algorithm catalog	Completed
Formulation II (problem statement)	Completed
Discovery architecture	Completed
Supporting formalization library	Completed
Part I theorem	**Next**

Current problem. Validity of an uncertainty set after data-dependent hyperparameter selection on calibration data; quantify miscoverage deviation; restore coverage by randomizing selection.

Completed. Formulation I · stabilized-algorithm catalog · Formulation II (problem statement) · supporting discovery infrastructure · advisor-facing documentation.

Next theorem. Prove Part I replace-one stability under explicit assumptions; then obtain advisor feedback.

Effort reconstruction (92 hours)

Workstream	Hours
Mathematical Formulation I	**20**
Catalog of stabilized algorithms and literature synthesis	**20**
Mathematical Formulation II	**10**
Discovery architecture and supporting infrastructure	**18**
Lean verification, certification, and audit	**8**
Documentation, repository organization, diagrams, and advisor-facing materials	**10**
Meetings, planning, and feedback integration	**6**
Total	**92**

Hours exclude the April–May prelude. Estimates are conservative reconstructions.

Major intellectual contributions

Formulation I — broad stability framework (20 hours)

Formulation I

Earlier June work (June 25 email; June 29 meeting) developed a broad framework connecting algorithmic stability for post-hoc selection; optimization-based selectors; primitive decompositions; stabilization via noise, randomization, or regularization; reduced data splitting; and downstream inference.

What was learned. Concrete algorithms share recurring patterns once decomposed into primitives—motivating a catalog-driven search rather than a one-off construction. The June 25 framework was *student-proposed* and not yet approved.

Stabilized Algorithm Catalog (20 hours)

A primary research contribution: structured extraction of algorithms that had already been stabilized in the literature, used to motivate Formulation I and later support Formulation II.

- Identified concrete selection procedures that had been stabilized, and extracted the original algorithm together with the selected object.
- Recorded the stability notion, assumptions, and the stabilization mechanism (noise, randomization, or regularization), including noise distributions where relevant.
- Decomposed each algorithm into primitive operations and compared utility costs.
- Recorded proof strategies and downstream inference or generalization guarantees.
- Searched for recurring structural patterns across examples.
- Converted the results into a structured database and LaTeX artifacts that informed the later optimization-selected UQ formulation.

Formulation II — optimization-selected uncertainty quantification (10 hours)

Formulation II

Narrower current problem: uncertainty sets from calibration data whose hyperparameter is selected by constrained optimization on the *same* data.

- Validity after data-dependent hyperparameter selection; miscoverage deviation.
- Part I: replace-one stability of $\hat{\theta}(\mathcal{D})$.
- Part II: diagnose post-selection coverage; restore coverage by randomizing selection.
- Retain the fixed-θ uncertainty set construction while changing *how* the hyperparameter is selected (realized set generally changes with θ).

Decomposition. Part I: replace-one stability of $\hat{\theta}(\mathcal{D})$. Part II: validity and miscoverage after data-dependent selection; restore coverage by randomizing selection (no recalibration of $\mathcal{C}$).

Supporting infrastructure

Discovery architecture and a modest formalization library support the mathematical program. They should enable—not replace—the Part I theorem.

Prelude: selecting the research pillar (April 27–May 22)

Focus. Select a lab research pillar connected to CREDO/CREAM.

What changed. Three candidate pillars; Wenbin entered the discussion; algorithmic pillar became active.

What was learned. High-stakes UQ: when can a data-produced decision be trusted?

Pivot. Quantization set aside; adaptive certified regret frontier became the plan. CREAM backups remained unused.

Prelude hours are not included in the 92-hour reconstruction.

Weekly intellectual timeline

Week 1: June 1–7 5 hours

Focus. Adaptive-regret continuation; re-entry after June 1.

Key developments. Reviewed materials; re-established advisor context. No new pivot documented.

Outputs. None retained beyond re-entry notes.

Evidence. *Weakly documented*; no undocumented progress inferred.

Week 2: June 8–14 7 hours

Focus. Nicholas–Wenbin contact; post-hoc inference named.

Key developments. June 11 meeting (logistics only in notes). "Post-hoc inference" added to June 12 SURA form (Wenbin). Framing shifted toward inferential validity after data-dependent procedures—not yet optimization-based selection.

Outputs. Updated SURA form terminology.

Week 3: June 15–21 9 hours

Focus. Early stabilized-algorithm catalog.

Key developments. Organized stability / randomization / regularization / post-selection examples into a structured catalog (not a reading list). Began LaTeX operator and mechanism representations. Shift from regret construction toward reusable stabilization patterns.

Outputs. Initial catalog entries and LaTeX sketches.

Evidence. Daily sequence *inferred from surviving artifacts* and the June 25 email.

Week 4: June 22–28 12 hours

Focus. Student-proposed Formulation I architecture.

Key developments. Expanded literature database; primitive decompositions; noise / regularization / randomization; drafted framework linking algorithm, data, primitives, and inference. June 25 email asked whether this was the correct direction.

Outputs. June 25 email; meeting prep for June 29. Status: *student-proposed.*

Week 5: June 29–July 5 14 hours

Focus. Central pivot: optimization-induced policies and decisions.

Key developments. June 29 presentation connected the catalog to the Woody–Wenbin–Ryan thread. Selected object reframed as optimization solution / policy; discontinuity as the stability obstacle. July 2 written feedback authorized narrower methodological exploration. Adaptive regret ceased to be active.

Outputs. Meeting notes; new formulation/context corpus begun.

Week 6: July 6–12 15 hours

Focus. Formulation II: specific UQ instance.

Key developments. Formalized calibration data $\rightarrow$ uncertainty set; same data $\rightarrow$ constrained hyperparameter selection. Separated Part I (replace-one stability of $\hat{\theta}(\mathcal{D})$) from Part II (validity / miscoverage deviation; randomizing selection). Explored randomizing selection while retaining the fixed-θ uncertainty set construction.

Outputs. Revised problem write-up; early discovery tooling.

Week 7: July 13–19 14 hours

Focus. Discovery workflow in support of the formulation.
Key developments. Refined Part I / Part II notation; separated speculative discovery from later checking; structured intermediate representations. Infrastructure was becoming broader than the current theorem required.
Outputs. Architecture visuals; intermediate representations.

Week 8: July 20–26 11 hours

Focus. Supporting formalization, audit, and freeze.
Key developments. Audited and froze supporting formal artifacts on July 26 (≈50 complete operators; 53 `LEAN_FULL` packages as of the later library audit). Lesson: formalization should support—not replace—Part I.
Outputs. Frozen artifact set (July 26).

Week 9: July 27–29 5 hours

Focus. Advisor-facing consolidation.
Key developments. Polished Formulation II; mentor summaries; repository navigation with the problem write-up as front door; this timeline.
Outputs. Advisor packet materials; return plan centered on Part I.

Major intellectual transitions

Transition	What changed	Why it mattered
Quantization → regret	Engineering quantization replaced by adaptive certified regret.	Shifted toward guarantees for decisions.
Regret → stabilized-algorithm catalog	Stabilized algorithms organized by selector, primitive, mechanism, proof, guarantee.	Empirical basis for reusable patterns.
Generic selection → optimization-induced decisions	Object became an optimization solution or policy; discontinuity central.	Connected catalog to advisor thread; concrete object.
Broad framework → UQ instance	Formulation I narrowed to Formulation II.	Tractable Part I / Part II agenda.
Tool building → theorem-first	Infrastructure recognized as support, not the primary result.	Next: prove Part I, then feedback.

Clarifications. June 25 framework was *student-proposed*. June 29 meeting and July 2 feedback drove the central pivot.

Current formulation

Research question. Study when an uncertainty set $\mathcal{C}(X;\mathcal{D},\theta)$ remains valid after its hyperparameter is selected by constrained optimization on the same calibration data $\mathcal{D}$; quantify validity or miscoverage deviation; restore coverage by randomizing the selection map while retaining the existing fixed-θ uncertainty set construction.
The map $\mathcal{C}(\,\cdot\,;\mathcal{D},\theta)$ for fixed, prespecified θ is unchanged. Changing the selected hyperparameter ($\hat{\theta}(\mathcal{D})$ or $\widetilde{\theta}(\mathcal{D};\xi)$) generally changes the realized set.

Part I. Replace-one stability of $\hat{\theta}(\mathcal{D})$ under explicit assumptions.

Part II. Validity and miscoverage after data-dependent selection; restore coverage by randomizing selection (no recalibration of $\mathcal{C}$). Part I informs diagnosis; goal (iii) modifies the selection map.

Immediate theorem target. Prove Part I for the motivating problem; obtain advisor feedback; only then extend.

Outputs

- Mathematical formulations (Formulation I; Formulation II with Part I / Part II).
- Stabilized-algorithm catalog and structured literature database.
- Discovery architecture and supporting formal artifacts.
- Problem write-up; diagrams; documentation; evidence packet; repository navigation.

Reflection and corrected research strategy

The most important accomplishment was reaching Formulation II with a concrete theorem agenda. The stabilized-algorithm catalog remains a major contribution. Supporting infrastructure was generalized earlier than necessary. Advisor communication should have been earlier and more consistent.

Corrected strategy. Theorem-first execution with regular advisor checkpoints; use the catalog and infrastructure selectively in service of the proof.

Current state

Completed

- Formulation I;
- stabilized-algorithm catalog;
- Formulation II (problem statement);
- supporting discovery infrastructure;
- repository and advisor-facing documentation.

Immediate milestone

Complete the Part I theorem.

Longer-term direction

- Characterize data-dependent-selection validity and miscoverage;
- develop Part II (including selection randomization);
- assess whether the results support a broader framework or paper.